# Slick Trace investigation report — case_0003

Generated 2026-09-14T07:36:02.786051+00:00 · reproducible investigation report

**What this document is not.** This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

## Finding

**No vessel in this scene can account for the observed slick.** No candidate release reproduced the observation above the absolute fit gate. The system returns no name rather than the least-bad one; a source vessel may have left the scene before acquisition.

## Scene

| Field | Value |
|---|---|
| Incident | case_0003 |
| Acquired (UTC) | 2026-09-10T04:30:00+00:00 |
| Sensor | synthetic SAR-like (VV proxy, single channel) |
| Data status | synthetic |
| Pixel spacing | 433.59 m |
| Label source | generator ground truth (synthetic) — not a human or independently confirmed label |

## Evidence layers

| Layer | Provenance |
|---|---|
| AIS | synthetic (fictional MMSIs; no real broadcast data) |
| Forcing | slicktrace_core.synth_generator constant+spatial field v0.2 |
| Licence | Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply. |

## Traffic filtering

| State | Vessels |
|---|---|
| excluded | 3 |
| candidate | 2 |

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

## Rank fragility

Leader retained in 0.7 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

## File integrity

| File | SHA-256 (first 16) |
|---|---|
| attribution_result.json | 182cfa47167af72f |
| coverage_result.json | 3d64094b061ffdd1 |
| forcing.npz | 767ec8f532e8cc46 |
| forecast_result.json | f02d0ac96b10581a |
| fragility_result.json | 251152903d437682 |
| historical_mapping.json | 39e1a2934072dc40 |
| manifest.json | ac936583396881e2 |
| mask_groundtruth.png | a9dedae51c4cf6cb |
| naive_baseline_result.json | fedf1188f395724a |
| next_observation_result.json | c12e2e32d9f6468a |
| scene.png | 4aec948e59525470 |

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

## Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

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